

EASTERN RAILWAY PCSTE DOCUMENT – HOWRAH SECTION

1. KAVACH – The Indian Railway ATP

Railway	ER	Division	HOWRAH
Section Name	HOWRAH-BARDHAMAN	Section details	94 RKM
No of stations	20 STATIONS	No of Mid-Section	6

Applicable RDSO specification	a)RDSO/SPN/196/2020 Version 4.0 b)TAN-1: RDSO Lr No. STS/E/TAN/5001, Ver. 1.0, dt: 20.02.2019 c)TAN-2: RDSO-SIG0TCAS(TAN)-1-2021 Ver. 2.0, dt: 11.10.2021 d) RDSO-SIG0TCAS(TAN)-1-2021 Ver.3.0, Dt : 12-03-2025
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2. Introduction

Purpose: To commission Kavach as per RDSO/SPN/196/2020 Version 4.0 in HOWRAH-BARDHAMAN section of ER Railway.

2.2 Reference Documents

S.No	Document Name	Reference/Details
1	Signal Interlocking Plan	Reference SIP is enclosed as Annexure I
2	Route Control Chart (RCC)	Reference Route Control Chart is enclosed as Annexure II
3	Gradient Profile Document	Reference Gradient Profile Document is enclosed as Annexure III
4	WTT for PSRs, manned LC Gates & Operating information for MPS.	Reference Howrah Division Working time table is enclosed as Annexure IV

2.3 Scope Of Work

"Provision of KAVACH alongwith 2x24 Fibre OFC Backbone in connection with work of raising of speed up to 160KMPH on Howrah - Pradhankhanta (Excl.) section of Howrah and Asansol Division of Eastern Railway"

2.3.1 Details of Sections

Section Name	Section Details	No. of Stations	No. of Mid-Sections	Applicable RDSO Spec
HOWRAH-BARDHAMAN	94 RKMs	20	06 LC gate	RDSO/SPN/196/2020 Version 4.0

2.3.2 Absolute Block Signalling system overview: Saktigarh To Bardhaman:Absolute Block Signalling

2.3.3 Automatic Block Signalling system overview: Howrah To Saktigarh:Auto Section Signalling

2.3.4 KAVACH Details

a) Make of KAVACH HBL

b) KAVACH Executive Checksums: As per RDSO approval letter No. RDSO-SIG0TCAS(HBL)/1/2025 Dated 13/10/2025

c)Station Kavach location and details of connected RIUs mentioned in the below table:

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Station Kavach Name	Station Code	Stationary Kavach ID	Location of Tower	Connected RIU	Executive Checksum
HOWRAH	HWH	12000	0.725	HWH RIU1, RIU2, RIU3, RIU4, RIU5, RIU6	2C77ECCA
HOWRAH	HWH	12000	0.725	HWH RIU1, RIU2, RIU3, RIU4, RIU5, RIU6	2C77ECCA
HOWRAH	HWH	12000	0.725	HWH RIU1, RIU2, RIU3, RIU4, RIU5, RIU6	2C77ECCA
HOWRAH	HWH	12000	0.725	HWH RIU1, RIU2, RIU3, RIU4, RIU5, RIU6	2C77ECCA
HOWRAH	HWH	12000	0.725	HWH RIU1, RIU2, RIU3, RIU4, RIU5, RIU6	2C77ECCA
HOWRAH	HWH	12000	0.725	HWH RIU1, RIU2, RIU3, RIU4, RIU5, RIU6	2C77ECCA
HOWRAH	HWH	12000	0.725	HWH RIU1, RIU2, RIU3, RIU4, RIU5, RIU6	2C77ECCA
HOWRAH	HWH	12000	0.725	HWH RIU1, RIU2, RIU3, RIU4, RIU5, RIU6	2C77ECCA
HOWRAH	HWH	12000	0.725	HWH RIU1, RIU2, RIU3, RIU4, RIU5, RIU6	2C77ECCA
HOWRAH	HWH	12000	0.725	HWH RIU1, RIU2, RIU3, RIU4, RIU5, RIU6	2C77ECCA

3 System Description

3.1 Signaling Description

3.1.1 Types of Signals, their Aspects: Multi Aspect Color Light signaling; Aspects as Per Signal Interlocking Plans (Annexure-I)

3.1.2 Signal Naming Rule: As per Signal Interlocking Plans (Annexure-I)

3.1.3 LC gate Naming Rule: As per Signal Interlocking Plans (Annexure-I)

3.1.4 Reference Documents

S.NO	Station Kavach Name	SIP No	RCC No	Working Time Table	ESP	Index Plan
1	HOWRAH	SI-07514/1 ALT-XXII	SI-07514/1 TO SUIT SIP ALT-VIII	2025	Y-596-2016 (ALT-I)	Enclosed Annexure-III (Index Plan)
2	HOWRAH	SI-07514/1 ALT-XXII	SI-07514/1 TO SUIT SIP ALT-VIII	2025	Y-596-2016 (ALT-I)	Enclosed Annexure-III (Index Plan)
3	HOWRAH	SI-07514/1 ALT-XXII	SI-07514/1 TO SUIT SIP ALT-VIII	2025	Y-596-2016 (ALT-I)	Enclosed Annexure-III (Index Plan)
4	HOWRAH	SI-07514/1 ALT-XXII	SI-07514/1 TO SUIT SIP ALT-VIII	2025	Y-596-2016 (ALT-I)	Enclosed Annexure-III (Index Plan)
5	HOWRAH	SI-07514/1 ALT-XXII	SI-07514/1 TO SUIT SIP ALT-VIII	2025	Y-596-2016 (ALT-I)	Enclosed Annexure-III (Index Plan)

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S.NO	Station Kavach Name	SIP No	RCC No	Working Time Table	ESP	Index Plan
6	HOWRAH	SI-07514/1 ALT-XXII	SI-07514/1 TO SUIT SIP ALT-VIII	2025	Y-596-2016 (ALT-I)	Enclosed Annexure-III (Index Plan)Enclosed Annexure-III (Index Plan)
7	HOWRAH	SI-07514/1 ALT-XXII	SI-07514/1 TO SUIT SIP ALT-VIII	2025	Y-596-2016 (ALT-I)	Enclosed Annexure-III (Index Plan)
8	HOWRAH	SI-07514/1 ALT-XXII	SI-07514/1 TO SUIT SIP ALT-VIII	2025	Y-596-2016 (ALT-I)	Enclosed Annexure-III (Index Plan)
9	HOWRAH	SI-07514/1 ALT-XXII	SI-07514/1 TO SUIT SIP ALT-VIII	2025	Y-596-2016 (ALT-I)	Enclosed Annexure-III (Index Plan)
10	HOWRAH	SI-07514/1 ALT-XXII	SI-07514/1 TO SUIT SIP ALT-VIII	2025	Y-596-2016 (ALT-I)	Enclosed Annexure-III (Index Plan)

3.1.5 Entry and Exit Signals, Direction of Travel their interpretation in sections of Movement Authority:

S.NO	Stationary Kavach ID	ENTRY Signal NAME	EXIT Signal NAME	ENTRY Signal ASPECT	EXIT Signal ASPECT	MAIN SECTIONS
Station Kavach TOC are enclosed as Annexure-VI						

3.2 Stationary KAVACH

3.2.1 Stationary Kavach ID Naming Rule:As per RDSO TAN number RDSO-SIG0TCAS(TAN)/1/2021 dated 18.10.2021

3.2.2 RFID Tag ID Naming Rule: As per RDSO/SPN/196/2020 Version 4.0 Annexure - H - Specification of KAVACH (The Indian Railway ATP)- RFID Tag-TIN Lay- Amdt-5 Para 2.23 (Enclosed TIN TAG Numbering Scheme as Annexure VII)

3.2.3 TIN ID Naming Rule: As per RDSO/SPN/196/2020 Version 4.0 Annexure - H -Specification of KAVACH (The Indian Railway ATP)- RFID Tag-TIN Lay- Amdt-5 Para 2.23 (Enclosed TIN TAG Numbering Scheme as Annexure VII)

3.2.4 Maximum number of Simultaneous Moving and Stationary trains:

Details are mentioned in the below table

3.2.5 Listing The Details

S.NO	Station Name	Station Code	Stationary Kavach ID	RFID Group ID	TIN Group ID	Maximum Simultaneous Movement	Maximum Stationary Trains
1	HOWRAH	HWH	12000	1-380	32874	12	21
2	HOWRAH	HWH	12000	1-380	32874	12	21
3	HOWRAH	HWH	12000	1-380	32874	12	21
4	HOWRAH	HWH	12000	1-380	32874	12	21
5	HOWRAH	HWH	12000	1-380	32874	12	21
6	HOWRAH	HWH	12000	1-380	32874	12	21

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S.NO	Station Name	Station Code	Stationary Kavach ID	RFID Group ID	TIN Group ID	Maximum Simultaneous Movement	Maximum Stationary Trains
7	HOWRAH	HWH	12000	1-380	32874	12	21
8	HOWRAH	HWH	12000	1-380	32874	12	21
9	HOWRAH	HWH	12000	1-380	32874	12	21
10	HOWRAH	HWH	12000	1-380	32874	12	21

3.2.6 RFID placement and Linking distances:

S.NO	Stationary Kavach ID	ENTRY Signal	EXIT Signal	REF RFID(Signal Foot Tag)	RFIDs in the Route and Linking Distances	Distance between Entry and Exit Signal
Details are available in Station Kavach TOC are enclosed as Annexure-VI						

3.2.7 SIM Cards details:

3.2.7.1 Station Kavach

S.NO	Station Name	Station Code	Primary SIM ID	Primary SIM Post Paid	Secondary SIM ID	Secondary SIM Post Paid
1	HOWRAH	HWH	9147923358	AIRTEL Postpaid	6290195358	JIO Postpaid
2	SORTING YARD	SYAE	9147923340	AIRTEL Postpaid	6290196095	JIO Postpaid
3	SORTING	SYAE	9147923340	AIRTEL Postpaid	6290196095	Postpaid
4	SORTING	SYAE	9147923340	AIRTEL Postpaid	6290196095	Postpaid
5	SORTING	SYAE	9147923340	AIRTEL Postpaid	6290196095	Postpaid
6	SORTING	SYAE	9147923340	AIRTEL Postpaid	6290196095	Postpaid
7	SORTING	SYAE	9147923340	AIRTEL Postpaid	6290196095	Postpaid
8	SORTING	SYAE	9147923340	AIRTEL Postpaid	6290196095	Postpaid
9	SORTING	SYAE	9147923340	AIRTEL Postpaid	6290196095	Postpaid
10	SORTING	SYAE	9147923340	AIRTEL Postpaid	6290196095	Postpaid

3.2.7.2 Onboard Kavach SIM Card Details:

S.NO	Shed	Loco ID	SIM - 1	Postpaid Operator	SIM-2	Postpaid Operator
1	EMU	13160	9875688858	Jio	9220102348	Airtel
2	EMU	13160	9875688858	Jio	9220102348	Airtel
3	EMU	13160	9875688858	Jio	9220102348	Airtel
4	EMU	13160	9875688858	Jio	9220102348	Airtel
5	EMU	13160	9875688858	Jio	9220102348	Airtel
6	EMU	13160	9875688858	Jio	9220102348	Airtel
7	EMU	13160	9875688858	Jio	9220102348	Airtel

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S.NO	Shed	Loco ID	SIM - 1	Postpaid Operator	SIM-2	Postpaid Operator
8	EMU	13160	9875688858	Jio	9220102348	Airtel
9	EMU	13160	9875688858	Jio	9220102348	Airtel
10	EMU	13160	9875688858	Jio	9220102348	Airtel

3.3 Track Profile Description

3.3.1 Centre line of Stations

S.NO	Station Kavach Name	Station Code	Center Line As per SIP	Center Line (As per RFID Layout)	STD III SIP No
1	HOWRAH	HWH	0.76585	RFID_TAG_LAYOUT_HWH_2.0.12 (AS MADE)	RFID_TAG_LAYOUT_HWH_2.0.12(AS MADE)
2	HOWRAH	HWH	0.76585	RFID_TAG_LAYOUT_HWH_2.0.12	RFID_TAG_LAYOUT_HWH_2.0.12
3	HOWRAH	HWH	0.76585	RFID_TAG_LAYOUT_HWH_2.0.12	RFID_TAG_LAYOUT_HWH_2.0.12
4	HOWRAH	HWH	0.76585	RFID_TAG_LAYOUT_HWH_2.0.12	RFID_TAG_LAYOUT_HWH_2.0.12
5	HOWRAH	HWH	0.76585	RFID_TAG_LAYOUT_HWH_2.0.12	RFID_TAG_LAYOUT_HWH_2.0.12
6	HOWRAH	HWH	0.76585	RFID_TAG_LAYOUT_HWH_2.0.12	RFID_TAG_LAYOUT_HWH_2.0.12
7	HOWRAH	HWH	0.76585	RFID_TAG_LAYOUT_HWH_2.0.12	RFID_TAG_LAYOUT_HWH_2.0.12
8	HOWRAH	HWH	0.76585	RFID_TAG_LAYOUT_HWH_2.0.12	RFID_TAG_LAYOUT_HWH_2.0.12
9	HOWRAH	HWH	0.76585	RFID_TAG_LAYOUT_HWH_2.0.12	RFID_TAG_LAYOUT_HWH_2.0.12
10	HOWRAH	HWH	0.76585	RFID_TAG_LAYOUT_HWH_2.0.12	RFID_TAG_LAYOUT_HWH_2.0.12

3.3.2 Maximum Permissible Speed

S.NO	Stationary Kavach ID	ENTRY Signal	EXIT Signal	REF RFID(Signal Foot Tag)	Distance from ref RFID	Location	MPS
Details as per the Track profile TOC Annexure IX (Along with ABS location of signal taken from concerned Station RFID Layout)							

3.3.3 Gradient Profile

S.No	Stationary Kavach ID	ENTRY Signal	EXIT Signal	REF RFID(Signal Foot Tag)	Distance from ref RFID	Location	Grad_Ty pe	Grad_Val ue	Grad_Le ngth	N_Valu e
Details as per the Track profile TOC Annexure IX (Along with ABS location of signal taken from concerned Station RFID Layout)										

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3.3.4 Turnout Profile

S.No	Stationary Kavach ID	ENTRY Signal	EXIT Signal	REF RFID (Signal Foot Tag)	Start Distance from ref RFID	Length	Turnout Speed
Details as per the Track profile TOC Annexure IX (Along with ABS location of signal taken from concerned Station RFID Layout)							

3.3.5 LC Gate Profile

S.NO	Stationary Kavach ID	ENTRY Signal	EXIT Signal	REF RFID(Signal Foot Tag)	Start Distance from ref RFID	LC gate Name	LC Gate Type
Details as per the Track profile TOC Annexure IX							

3.3.6 OHE mast or mileage Posts Profile

S.No	Stationary Kavach ID	ENTRY Signal	EXIT Signal	REF RFID(Signal Foot Tag)	Distance from ref RFID	Location	OHE Mast/mileage post details
Not Applicable							

3.3.7 TSR Route IDs

S.No	Stationary Kavach ID	ENTRY Signal	EXIT Signal	TSR Route IDs	TSR Route ID Start distance
Not Applicable					

3.3.8 Section type details

S.No	Stationary Kavach ID	ENTRY Signal	EXIT Signal	Section type	Shunting Limit Tag or not	Communication required in Nominal or not	Communication required in Reverse or not
Details are as per the TOC Annexure VI Required & provided at all locations Required & provided at all locations							

3.3.9 Track Description Details

S.No	Stationary Kavach ID	ENTRY Signal	EXIT Signal	Distance from ref RFID	Track Description	Start Location	Length
Details are as per the TOC Annexure VI							

3.3.10 Adjustment Tag/ Junction Tag Details:

S.No	Stationary Kavach ID	ENTRY Signal	EXIT Signal	A-Tag RFID NO	ABS-1	ABS-2	Location Adjustemnt	Junctio n Adjust ment	Hazards Identifi ed	Mitigation Measure
Adjustment Tag/ Junction Tags are not used in the HWH-BWN section.										

4 National Values

4.1 Stationary KAVACH Configurable Parameters:

4.1.1 Station related

S.No	Station	Type	Shunt	Limit	Shunt Limit	IP	Port 1	Port 2
Details are as per the TOC Annexure VII Details of IP Address and port number are available in Annexure X KAVACH Approved IP Plan								

4.1.2 Radio Modem:

Power, Frequency Resolution, Control frequency, transmitting frequency, receiving frequency

a)Radio Modem Power: 10 Watt

b)Frequency Resolution: 1KHZ

c)Control frequency: 427.625 MHZ

d)Transmitting frequency: Details are available in "Annexure XI KAVACH Frequency Plan and Time Slot allotment chart.

e)Receiving frequency: Details are available in "Annexure XI KAVACH Frequency Plan and Time Slot allotment chart.

4.1.3 Time Slot Management: Details are available in "Annexure XI KAVACH Frequency Plan and Time Slot allotment chart.

4.1.4 Timeout parameters

Communication Timeout: Refer Sr No:2.3 ,2.4 & 2.5 in the below table

Deregistration Timeout: Refer Sr No:2.6 ,2.7 in the below table

4.1.5 Time Related: Refer Sr No: 1.8,1.9,1.18,2.9 & 2.15 in the below table

4.1.6 Distance Related: Refer Sr No:2.10 ,2.11 & 2.13 in the below table

4.1.7 Sps Related: Refer to Sr. No 2.9 & 2.12 in the below table

4.1.8 Data Logging:

a)Detailed data logging: 240 hours

b)Maintenance data logging: 90 days

c)Critical fault data: 180 days

4.1.9 GSM APN details: Details are available in the below table: JIO 4G postpaid are provided by Railways, Details available in Para 3.2.7

4.2.1 Loco Related

S.No	Loco ID	Loco Model	Loco brake Interface	Wheel Dia1	Wheel Dia2	Max speed	Loco Fsb	Loco Eb
1	30082	WAP5	E70	1077mm	1077mm	160km/h	1.0 m/s ²	1.5 m/s ²
2	30082	WAP5	E70	1077mm	1077mm	160km	1.0 m/s ²	1.5 m/s ²
3	30082	WAP5	E70	1077mm	1077mm	160km	1.0 m/s ²	1.5 m/s ²
4	30082	WAP5	E70	1077mm	1077mm	160km	1.0 m/s ²	1.5 m/s ²
5	30082	WAP5	E70	1077mm	1077mm	160km	1.0 m/s ²	1.5 m/s ²
6	30082	WAP5	E70	1077mm	1077mm	160km	1.0 m/s ²	1.5 m/s ²
7	30082	WAP5	E70	1077mm	1077mm	160km	1.0 m/s ²	1.5 m/s ²
8	30082	WAP5	E70	1077mm	1077mm	160km	1.0 m/s ²	1.5 m/s ²
9	30082	WAP5	E70	1077mm	1077mm	160km	1.0 m/s ²	1.5 m/s ²
10	30082	WAP5	E70	1077mm	1077mm	160km	1.0 m/s ²	1.5 m/s ²

4.2.2 Radio Modem

Power, Frequency Resolution, Control Frequency, transmitting frequency, receiving Frequency

a. Radio Modem Power: 10 Watt

b. Frequency Resolution: 1KHZ

c. Control frequency: 427.625

d. Transmitting frequency: Details are available in "Annexure XI KAVACH Frequency Plan and Time Slot allotment chart.

e. Receiving frequency: Details are available in "Annexure XI KAVACH Frequency Plan and Time Slot allotment chart.

4.2.3 RFID Readers

Reader1 offset from front end and back end, Reader2 offset from frontend and backend, Location Accuracy, L_Doubtlover, L_Doubtunder

a. Reader1 offset from front end: 2mts

b. Reader1 offset from back end: 18mts

c. Reader2 offset from front end: 2mts

d. Reader2 offset from back end: 18mts

e. Location Accuracy : 5mts

f. L_Doubtlover : 5%

g. L_Doubtunder .5%

4.2.4 Speed sensor related

Tacho Pulses/Rev, Tacho type and its mounting direction.

a. Tacho Pulses/Rev, Tacho type and its mounting direction.

b. Tacho Type: Quadrature

c. Tacho Mount Dir for SPD1: Left

d. Tacho Mount Dir for SPD2: Right

4.2.5 Speed Margins

a. NB: 5 Kmph

b. FSB: 8 Kmph

c. EB: 10 Kmph

4.2.6 Margin Distances

a.Speed margin warning: 2 Kmph

4.2.7 LP Reaction Time

7 Sec

4.2.8 Time Slot Management

Time slots are allotted to Loco by Station Kavach, thePlanning of these time slots is available in Approved Frequency Plan (Enclosed as Annexure:XI)

4.2.9 Timeout Parameters

Communication Timeout: 30 seconds in absolute section and 10 seconds in automatic section
Override Timeout: 240 sec

4.2.10 SoS related

- a.SOS Trigger distance:3000 mts
- b.SOS speed: 30 kmph
- c.SOS cancel distance:1500mts
- d.SOS Timeout: 180 days

4.2.11 Data Logging

- a.Detailed data logging: 240 hours
- b.Maintenance data logging :90 days
- c.Critical fault data : 180 days

4.2.12 GSM APN details

Airtel & Jio Postpaid are provided by Railways

4.2.13 Fault Codes

Write manually

4.3 TSRMS Configurable Parameters -

Not Applicable

4.3.1 Sections, Stations in section, RE/Non-RE:

Not Applicable

4.3.2 OHE Pole or mileage post, RFID, Distance from RFID, TSR Route ID and Stationary Kavach ID:

Not Applicable

4.4 NMS Configurable Parameters enclosed in Section Common documents

4.4.1 Prompt requirements

Written manually

4.4.2 SMS requirements

Written manually

4.4.3 Display requirements

The HBL NMS application is implemented as per Annexure-G of Kavach Specification document.
[(Annexure-G-Specification of Kavach RDSO/SPN/196/2020 Version 4.0) The Indian Railway ATP)-
Network Monitoring System Protocol-Amdt-4 16_07_2024]

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4.4.4 Report requirements

written manually

5 Site installation

5.1 RFID Tags

5.1.1 RFID Tag Data Validation Report – Enclosed as Annexure XII A

5.1.2 RFID Tag Placement Verification Report – Enclosed as Annexure XII B

5.2 Radio

5.2.1 Towers

S.NO	Stationary Kavach ID	Tower height in metres	Soil test Report	Approved Foundation Drawing	Approved Tower Drawing	Approval From Division	SACFA Clearance
1	12000 HOWRAH	40 Mtr	10.30 t/m	Drg. No. 2021-PACS-SPIRE-407 RH-40M-100KPA-GBT-WT-FDN-001 21-12-21	Drg. No. RDSO/TC/TOWER/L/40/180/T/6.5	Division Approved General Arrangement are enclosed	DL letter issued by WPC vide Letter No. NW D317202310 16324 Date: 20-02-2024
2	12000 HOWRAH	40 Mtr	10.30 t/m	Drg. No. 2021-PACS-SPIRE-407 RH-40M-100KPA-GBT-WT-FDN-001 21-12-21	Drg. No. RDSO/TC/TOWER/L/40/180/T/6.5	Division Approved General Arrangement are enclosed	DL letter issued by WPC vide Letter No. NW D317202310 16324 Date: 20-02-2024
3	12000 HOWRAH	40 Mtr	10.30 t/m	Drg. No. 2021-PACS-SPIRE-407 RH-40M-100KPA-GBT-WT-FDN-001 21-12-21	Drg. No. RDSO/TC/TOWER/L/40/180/T/6.5	Division Approved General Arrangement are enclosed	DL letter issued by WPC vide Letter No. NW D317202310 16324 Date: 20-02-2024
4	12000 HOWRAH	40 Mtr	10.30 t/m	Drg. No. 2021-PACS-SPIRE-407 RH-40M-100KPA-GBT-WT-FDN-001 21-12-21	Drg. No. RDSO/TC/TOWER/L/40/180/T/6.5	Division Approved General Arrangement are enclosed	DL letter issued by WPC vide Letter No. NW D317202310 16324 Date: 20-02-2024
5	12000 HOWRAH	40 Mtr	10.30 t/m	Drg. No. 2021-PACS-SPIRE-407 RH-40M-100KPA-GBT-WT-FDN-001 21-12-21	Drg. No. RDSO/TC/TOWER/L/40/180/T/6.5	Division Approved General Arrangement are enclosed	DL letter issued by WPC vide Letter No. NW D317202310 16324 Date: 20-02-2024
6	12000 HOWRAH	40 Mtr	10.30 t/m	Drg. No. 2021-PACS-SPIRE-407 RH-40M-100KPA-GBT-WT-FDN-001 21-12-21	Drg. No. RDSO/TC/TOWER/L/40/180/T/6.5	Division Approved General Arrangement are enclosed	DL letter issued by WPC vide Letter No. NW D317202310 16324 Date: 20-02-2024

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S.NO	Stationary Kavach ID	Tower height in metres	Soil test Report	Approved Foundation Drawing	Approved Tower Drawing	Approval From Division	SACFA Clearance
7	12000 HOWRAH	40 Mtr	10.30 t/m	Drg. No. 2021-PACS-SPIRE-407 RH-40M-100KPA-GBT-WT-FDN-001 21-12-21	Drg. No. RDSO/TC/ TOWER/ L/40/180/ T/6.5	Division Approved General Arrangement are enclosed	DL letter issued by WPC vide Letter No. NW D317202310 16324 Date: 20-02-2024
8	12000 HOWRAH	40 Mtr	10.30 t/m	Drg. No. 2021-PACS-SPIRE-407 RH-40M-100KPA-GBT-WT-FDN-001 21-12-21	Drg. No. RDSO/TC/ TOWER/ L/40/180/ T/6.5	Division Approved General Arrangement are enclosed	DL letter issued by WPC vide Letter No. NW D317202310 16324 Date: 20-02-2024
9	12000 HOWRAH	40 Mtr	10.30 t/m	Drg. No. 2021-PACS-SPIRE-407 RH-40M-100KPA-GBT-WT-FDN-001 21-12-21	Drg. No. RDSO/TC/ TOWER/ L/40/180/ T/6.5	Division Approved General Arrangement are enclosed	DL letter issued by WPC vide Letter No. NW D317202310 16324 Date: 20-02-2024
10	12000 HOWRAH	40 Mtr	10.30 t/m	Drg. No. 2021-PACS-SPIRE-407 RH-40M-100KPA-GBT-WT-FDN-001 21-12-21	Drg. No. RDSO/TC/ TOWER/ L/40/180/ T/6.5	Division Approved General Arrangement are enclosed	DL letter issued by WPC vide Letter No. NW D317202310 16324 Date: 20-02-2024

5.2.2 Communication coverage Report

- Communication charts showing coverage up to the communication mandatory area for transmission and reception of control and operational frequency shall be submitted.
- The summary shall be as follows:

5.2.2.1 Transmission side (Control Frequency)

S.No	Stationary Kavach Id	Tx Communication (Up)	Tx Communication (DN)	Tx Frequency (MHz)	Tx Frequency (Up)	Tx Frequency (DN)	Is distance more than Comm
1	12000	0.019	2.311	425.575	0.706	1.586	Yes
2	12000	0.019	2.311	425.575	0.706	1.586	Yes
3	12000	0.019	2.311	425.575	0.706	1.586	Yes
4	12000	0.019	2.311	425.575	0.706	1.586	Yes
5	12000	0.019	2.311	425.575	0.706	1.586	Yes
6	12000	0.019	2.311	425.575	0.706	1.586	Yes
7	12000	0.019	2.311	425.575	0.706	1.586	Yes
8	12000	0.019	2.311	425.575	0.706	1.586	Yes
9	12000	0.019	2.311	425.575	0.706	1.586	Yes

EASTERN RAILWAY PCSTE DOCUMENT – HOWRAH SECTION

S.No	Stationary Kavach Id	Tx Communication (Up)	Tx Communication (DN)	Tx Frequency(MHz)	Tx Frequency (Up)	Tx Frequency (DN)	Is distance more than Comm
10	12000	0.019	2.311	425.575	0.706	1.586	Yes

5.2.2.2 Transmission side (Operational Frequency)

S.No	Stationary Kavach Id	Tx Communication (Up)	Tx Communication (DN)	Tx Frequency(MHz)	Tx Frequency (Up)	Tx Frequency (DN)	Is distance more than Comm
1	12000	0.019	2.311	425.575	0.706	1.586	Yes
2	12000	0.019	2.311	425.575	0.706	1.586	Yes
3	12000	0.019	2.311	425.575	0.706	1.586	Yes
4	12000	0.019	2.311	425.575	0.706	1.586	Yes
5	12000	0.019	2.311	425.575	0.706	1.586	Yes
6	12000	0.019	2.311	425.575	0.706	1.586	Yes
7	12000	0.019	2.311	425.575	0.706	1.586	Yes
8	12000	0.019	2.311	425.575	0.706	1.586	Yes
9	12000	0.019	2.311	425.575	0.706	1.586	Yes
10	12000	0.019	2.311	425.575	0.706	1.586	Yes

5.2.2.3 Reception side (Control Frequency)

S.No	Stationary Kavach Id	Tx Communication (Up)	Tx Communication (DN)	Tx Frequency(MHz)	Tx Frequency (Up)	Tx Frequency (DN)	Is distance more than Comm
1	12000	0.019	2.311	425.575	0.706	1.586	Yes
2	12000	0.019	2.311	425.575	0.706	1.586	Yes
3	12000	0.019	2.311	425.575	0.706	1.586	Yes
4	12000	0.019	2.311	425.575	0.706	1.586	Yes
5	12000	0.019	2.311	425.575	0.706	1.586	Yes
6	12000	0.019	2.311	425.575	0.706	1.586	Yes
7	12000	0.019	2.311	425.575	0.706	1.586	Yes
8	12000	0.019	2.311	425.575	0.706	1.586	Yes
9	12000	0.019	2.311	425.575	0.706	1.586	Yes
10	12000	0.019	2.311	425.575	0.706	1.586	Yes

EASTERN RAILWAY PCSTE DOCUMENT – HOWRAH SECTION

5.2.2.4 Reception side (Operational Frequency)

S.No	Stationary Kavach Id	Rx Communication (Up)	Rx Communication (DN)	Rx Frequency (MHz)	Rx Frequency (Up)	Rx Frequency (DN)	Is distance more than Comm
1	12000	0.019	2.311	427.875	0.706	1.586	Yes
2	12000	0.019	2.311	427.875	0.706	1.586	Yes
3	12000	0.019	2.311	427.875	0.706	1.586	Yes
4	12000	0.019	2.311	427.875	0.706	1.586	Yes
5	12000	0.019	2.311	427.875	0.706	1.586	Yes
6	12000	0.019	2.311	427.875	0.706	1.586	Yes
7	12000	0.019	2.311	427.875	0.706	1.586	Yes
8	12000	0.019	2.311	427.875	0.706	1.586	Yes
9	12000	0.019	2.311	427.875	0.706	1.586	Yes
10	12000	0.019	2.311	427.875	0.706	1.586	Yes

5.2.3 Time Slot allotment chart :

- Maximum trains that can be handled by Stationary KAVACH shall be 28.
- In case of exceptional circumstances, the main lines and connected lines from main line shall be given preference.

S.No	Stationary Kavach Id	Max no of simultaneous movements and associated Packet Length from station	Max no of Stationary movements	Total time slots required	Time slots allotted(<28)
Not Applicable					

5.3 Signaling

5.3.1 Relay wiring diagrams enclosed in Station Specific documents

S.No	Stationary Kavach ID	Prepared by	Checked by	1st level testing at site by	2nd level testing at site by	Approved by	Integrated with Station circuits or not
1	12000	V.Narasimha	Md.Rishad	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes(Ver.2.0.5)
2	12000	V.Narasimha	Md.Rishad	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes(Ver.2.0.5)
3	12000	V.Narasimha	Md.Rishad	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes(Ver.2.0.5)
4	12000	V.Narasimha	Md.Rishad	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes(Ver.2.0.5)
5	12000	V.Narasimha	Md.Rishad	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes(Ver.2.0.5)
6	12000	V.Narasimha	Md.Rishad	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes(Ver.2.0.5)

EASTERN RAILWAY PCSTE DOCUMENT – HOWRAH SECTION

S.No	Stationary Kavach ID	Prepared by	Checked by	1st level testing at site by	2nd level testing at site by	Approved by	Integrated with Station circuits or not
7	12000	V.Narasimha	Md.Rishad	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes(Ver.2.0.5)
8	12000	V.Narasimha	Md.Rishad	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes(Ver.2.0.5)
9	12000	V.Narasimha	Md.Rishad	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes(Ver.2.0.5)
10	12000	V.Narasimha	Md.Rishad	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes(Ver.2.0.5)

5.3.2 Contact Analysis Charts enclosed in Station Specific documents

S.No	Stationary Kavach ID	Prepared by	Checked by	1st level testing at site by	2nd level testing at site by	Approved by	Integrated with Station circuits or not
1	12000	Narasimha Reddy	K.Chetan	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes
2	12000	Narasimha Reddy	K.Chetan	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes
3	12000	Narasimha Reddy	K.Chetan	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes
4	12000	Narasimha Reddy	K.Chetan	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes
5	12000	Narasimha Reddy	K.Chetan	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes
6	12000	Narasimha Reddy	K.Chetan	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes
7	12000	Narasimha Reddy	K.Chetan	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes
8	12000	Narasimha Reddy	K.Chetan	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes
9	12000	Narasimha Reddy	K.Chetan	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes
10	12000	Narasimha Reddy	K.Chetan	ASTE/W/RR/HQ	DY.CSTE/TELE/HQ	DY.CSTE/W/SDAH	Yes

5.3.3 Bit Charts enclosed in Station Specific documents

S.No	Stationary Kavach ID	Prepared by	Checked by	1st level testing at site by	2nd level testing at site by	Approved by
Auto generated bit chart. Jointly signed Bell Testing Report & Integrity Test Report. Enclosed in Annexure-XIX						

5.4 Point wise compliance to TAN issued by RDSO:

(Enclosed Annexure XIXA)

5.5 KAVACH Related

EASTERN RAILWAY PCSTE DOCUMENT – HOWRAH SECTION

5.5.1 Utilization of Input Boards:

S.No	Station name	Station Code	Stationary Kavach ID	Total No. of scope of bits as per hardware	Total No. of bits utilized as per hardware	Total No. of Spare Bits in absolute quantity	Total No. of Spare Bits in percentage
1	HOWRAH	HWH	12000	480	448	32	6.7%
2	HOWRAH	HWH	12000	480	448	32	6.7%
3	HOWRAH	HWH	12000	480	448	32	6.7%
4	HOWRAH	HWH	12000	480	448	32	6.7%
5	HOWRAH	HWH	12000	480	448	32	6.7%
6	HOWRAH	HWH	12000	480	448	32	6.7%
7	HOWRAH	HWH	12000	480	448	32	6.7%
8	HOWRAH	HWH	12000	480	448	32	6.7%
9	HOWRAH	HWH	12000	480	448	32	6.7%
10	HOWRAH	HWH	12000	480	448	32	6.7%

5.5.2 RFID Tag layouts

SN	Stationary Kavach ID	Drawing No	Prepared by	Checked by	Approved by
1	12000	RFID_TAG_LAYOUT_HWH_2.0.12(AS MADE)	Md.Rishad	K.Nikhitha	Dy.CSTE/W/SDAH
2	12000	RFID_TAG_LAYOUT_HWH_2.0.12(AS MADE)	Md.Rishad	K.Nikhitha	Dy.CSTE/W/SDAH
3	12000	RFID_TAG_LAYOUT_HWH_2.0.12(AS MADE)	Md.Rishad	K.Nikhitha	Dy.CSTE/W/SDAH
4	12000	RFID_TAG_LAYOUT_HWH_2.0.12(AS MADE)	Md.Rishad	K.Nikhitha	Dy.CSTE/W/SDAH
5	12000	RFID_TAG_LAYOUT_HWH_2.0.12(AS MADE)	Md.Rishad	K.Nikhitha	Dy.CSTE/W/SDAH
6	12000	RFID_TAG_LAYOUT_HWH_2.0.12(AS MADE)	Md.Rishad	K.Nikhitha	Dy.CSTE/W/SDAH
7	12000	RFID_TAG_LAYOUT_HWH_2.0.12(AS MADE)	Md.Rishad	K.Nikhitha	Dy.CSTE/W/SDAH
8	12000	RFID_TAG_LAYOUT_HWH_2.0.12(AS MADE)	Md.Rishad	K.Nikhitha	Dy.CSTE/W/SDAH
9	12000	RFID_TAG_LAYOUT_HWH_2.0.12(AS MADE)	Md.Rishad	K.Nikhitha	Dy.CSTE/W/SDAH
10	12000	RFID_TAG_LAYOUT_HWH_2.0.12(AS MADE)	Md.Rishad	K.Nikhitha	Dy.CSTE/W/SDAH

EASTERN RAILWAY PCSTE DOCUMENT – HOWRAH SECTION

5.5.3 RFID Tag Data Enclosed in Station Specific documents

SN	Stationary Kavach ID	Drawing No	Prepared by	Checked by	Approved by
1	12000	2.0.12	K.Umadevi	Md.Rishad	Dy.CSTE/W/SDAH
2	12000	2.0.12	K.Umadevi	Md.Rishad	Dy.CSTE/W/SDAH
3	12000	2.0.12	K.Umadevi	Md.Rishad	Dy.CSTE/W/SDAH
4	12000	2.0.12	K.Umadevi	Md.Rishad	Dy.CSTE/W/SDAH
5	12000	2.0.12	K.Umadevi	Md.Rishad	Dy.CSTE/W/SDAH
6	12000	2.0.12	K.Umadevi	Md.Rishad	Dy.CSTE/W/SDAH
7	12000	2.0.12	K.Umadevi	Md.Rishad	Dy.CSTE/W/SDAH
8	12000	2.0.12	K.Umadevi	Md.Rishad	Dy.CSTE/W/SDAH
9	12000	2.0.12	K.Umadevi	Md.Rishad	Dy.CSTE/W/SDAH
10	12000	2.0.12	K.Umadevi	Md.Rishad	Dy.CSTE/W/SDAH

5.5.4 KAVACH Table of Control Enclosed in Station Specific documents

SN	Stationary Kavach ID	Drawing No	Prepared by	Checked by	Approved by	FAT by	SAT by
1	12000	2.0.3	Nikhitha	B.Koteswar Rao	Dy.CSTE/W/SDAH	HBL & JE/SIG/W/HQ	SSE/SIG/W/HWH
2	12000	2.0.3	Nikhitha	B.Koteswar Rao	Dy.CSTE/W/SDAH	HBL & JE/SIG/W/HQ	SSE/SIG/W/HWH
3	12000	2.0.3	Nikhitha	B.Koteswar Rao	Dy.CSTE/W/SDAH	HBL & JE/SIG/W/HQ	SSE/SIG/W/HWH
4	12000	2.0.3	Nikhitha	B.Koteswar Rao	Dy.CSTE/W/SDAH	HBL & JE/SIG/W/HQ	SSE/SIG/W/HWH
5	12000	2.0.3	Nikhitha	B.Koteswar Rao	Dy.CSTE/W/SDAH	HBL & JE/SIG/W/HQ	SSE/SIG/W/HWH
6	12000	2.0.3	Nikhitha	B.Koteswar Rao	Dy.CSTE/W/SDAH	HBL & JE/SIG/W/HQ	SSE/SIG/W/HWH
7	12000	2.0.3	Nikhitha	B.Koteswar Rao	Dy.CSTE/W/SDAH	HBL & JE/SIG/W/HQ	SSE/SIG/W/HWH
8	12000	2.0.3	Nikhitha	B.Koteswar Rao	Dy.CSTE/W/SDAH	HBL & JE/SIG/W/HQ	SSE/SIG/W/HWH
9	12000	2.0.3	Nikhitha	B.Koteswar Rao	Dy.CSTE/W/SDAH	HBL & JE/SIG/W/HQ	SSE/SIG/W/HWH
10	12000	2.0.3	Nikhitha	B.Koteswar Rao	Dy.CSTE/W/SDAH	HBL & JE/SIG/W/HQ	SSE/SIG/W/HWH

5.5.5 KAVACH Track Profile Table Enclosed in Station Specific documents

SN	Stationary Kavach ID	Drawing No	Prepared by	Checked by	Approved by	FAT by	SAT by
1	12000	2.0.2	K.Nikhitha	JVV.SNV.Prasad	Dy.CSTE/W/SDAH	HBL & JE/SIG/W/HQ	SSE/SIG/W/HWH
2	12001	2.0.2	K.Nikhitha	B.Koteswara Rao	Dy.CSTE/W/SDAH	HBL & JE/SIG/W/HQ	SSE/SIG/W/HWH

EASTERN RAILWAY PCSTE DOCUMENT – HOWRAH SECTION

SN	Stationary Kavach ID	Drawing No	Prepared by	Checked by	Approved by	FAT by	SAT by
3	12001	2.0.2	K.Nikhitha	B.Koteswara Rao	B.Koteswara Rao	HBL & JE/ SIG/W/HQ	SSE/SIG/W/ HWH
4	12001	2.0.2	K.Nikhitha	B.Koteswara Rao	B.Koteswara Rao	HBL & JE/ SIG/W/HQ	SSE/SIG/W/ HWH
5	12001	2.0.2	K.Nikhitha	B.Koteswara Rao	B.Koteswara Rao	HBL & JE/ SIG/W/HQ	SSE/SIG/W/ HWH
6	12001	2.0.2	K.Nikhitha	B.Koteswara Rao	B.Koteswara Rao	HBL & JE/ SIG/W/HQ	SSE/SIG/W/ HWH
7	12001	2.0.2	K.Nikhitha	B.Koteswara Rao	B.Koteswara Rao	HBL & JE/ SIG/W/HQ	SSE/SIG/W/ HWH
8	12001	2.0.2	K.Nikhitha	B.Koteswara Rao	B.Koteswara Rao	HBL & JE/ SIG/W/HQ	SSE/SIG/W/ HWH
9	12001	2.0.2	K.Nikhitha	vB.Koteswara Rao	B.Koteswara Rao	HBL & JE/ SIG/W/HQ	SSE/SIG/W/ HWH
10	12001	2.0.2	K.Nikhitha	B.Koteswara Rao	B.Koteswara Rao	HBL & JE/ SIG/W/HQ	SSE/SIG/W/ HWH

5.5.6 KAVACH Table of Control (TSL)

SN	Stationary Kavach ID	Drawing No	Prepared by	Checked by	FAT by	Approved by	SAT by
Not applicable							

5.5.7 KAVACH Track Profile Table (TSL)

SN	Stationary Kavach ID	Drawing No	Prepared by	Checked by	FAT by	Approved by	SAT by
Not Applicable							

5.6 Test Reports

5.6.1 KAVACH TSR speed control test reports

S.No	Stationary Kavach ID	Route ID	Imposed Location by TSRMS	TSR Authorized by Onboard Kavach	SAT By	Is any Safety infringement observed?	Is any capacity Loss observed?
Not Applicable							

5.6.2 KAVACH PSR speed control test reports

S.No	Stationary Kavach ID	Entry Signal	PSR	Start Distance	SAT By	Is any safety infringement observed?	Is any capacity Loss observed/
(Enclosed Annexure-XX)							

EASTERN RAILWAY PCSTE DOCUMENT – HOWRAH SECTION

5.6.3 KAVACH SPAD Prevention test reports Enclosed in Station Specific documents

S.No	Stationary Kavach ID	Entry Signal	Exit Signal	Stopping distance from Exit Signal	SAT by	Is any Safety infringement observed?	Is any capacity Loss observed?
(Enclosed Annexure-XXI)							

5.6.4 KAVACH Rear End collision test reports

S.No	Stationary Kavach ID	Entry Signal	Exit Signal	Stopping Distance	SAT By	Is any Safety infringement observed?	Is any capacity loss observed?
(Enclosed Annexure-XXII)							

5.6.5 KAVACH Head ON collision test report

S.No	Stationary Kavach ID	Entry Signal	Exit Signal	Stopping Distance	SAT By	Is any Safety infringement observed?	Is any capacity loss observed?
(Enclosed Annexure-XXIII)							

5.6.6 KAVACH Unusual Stoppage in Block section test reports

S.No	Stationary Kavach ID	Entry Signal	Exit Signal	Stopping Distance	SAT By	Is any Safety infringement observed?	Is any capacity loss observed?
(Enclosed Annexure-XXIV)							

5.6.7 Manual SoS Generation from Stationary KAVACH test reports

S.No	Stationary Kavach ID	Entry Signal	Exit Signal	Stopping Distance	SAT By	Is any Safety infringement observed?	Is any capacity loss observed?
(Enclosed Annexure-XXV)							

5.6.8 Manual SoS Generation from Onboard KAVACH test reports

S.No	Stationary Kavach ID	Entry Signal	Exit Signal	Stopping Distance	SAT By	Is any Safety infringement observed?	Is any capacity loss observed?
(Enclosed Annexure-XVI)							

5.6.9 NMS evidences for all the above tests : Enclosed Annexure-XXVII

5.6.10 Staff training

Other Documentation Concerned Branch officer of division from S&T, Operating, TRSO, TRD, Engineering shall certify that their staff in the section are adequately trained for Kavach

EASTERN RAILWAY PCSTE DOCUMENT – HOWRAH SECTION

Operations and Maintenance.

(Enclosed Annexure- XXVIII)

6 Other Documentation

6.1 Stationary Kavach

6.1.1 SWR Amendments with introduction of Kavach SMOCIP operation : To be Provided By ER (Annexure XXIX)

6.1.2 FAT reports and Certificates - Enclosed as FAT Reports and certificate as Annexure XXX

6.1.3 SAT reports – Enclosed Annexure XXXI

6.1.4 Jointly Signed pre commissioning check list - Enclosed Annexure- XXXII

6.1.5 OEM Installation certificate - Enclosed Annexure-XXXIII

6.1.6 Data logger Validation Report for Kavach Relays – WIP, Relay wiring and cable termination completed. Data logger wiring and validation are pending. TDC 15-Sep-25 (Annexure XXXIV)

6.1.7 Meggaring/OTDR details of cables

Enclosed Reports Annexure XXXV)

6.1.8 Adequate Power Supply Planning Reports :

- Stationary Kavach: Enclosed in Station Specific documents
- Remote Interface Units: Enclosed in Station Specific documents
- Tower Locations: Enclosed in Station Specific documents
- SM OCIP : Enclosed in Station Specific documents
- TSRMS- --Not Applicable--

6.1.9 KMS Registration details - Enclosed Annexure XXXVII

6.1.10 Supply of adequate loco simulators and installation of stationary Kavach test benches.

- 1)2 Nos Loco simulators are available at site for the SAT testing
- 2)2 Test benches are supplied at HWH & SDAH sheds

6.1.11 Management of spares – Spares planned for maintenance are being dispatched on staggered manner, Spares details Enclosed Annexure XXXVIIB

6.1.12 Maintenance staff nomination and posting - Enclosed organization chart Annexure XXXVIIC

6.2 Onboard Kavach

6.2.1 About Locos

S.No.	Loco ID	Type of Loco	Type of BIU	Loco Shed
1	37005	WAP7	E70	HWH
2	37005	WAP7	E70	HWH
3	37005	WAP7	E70	HWH
4	37005	WAP7	E70	HWH
5	37005	WAP7	E70	HWH
6	37005	WAP7	E70	HWH

EASTERN RAILWAY PCSTE DOCUMENT – HOWRAH SECTION

S.No.	Loco ID	Type of Loco	Type of BIU	Loco Shed
7	37005	WAP7	E70	HWH
8	37005	WAP7	E70	HWH
9	37005	WAP7	E70	HWH
10	37005	WAP7	E70	HWH

6.2.2 Product Test certificates

- Inspection certificates of Onboard Kavach:
Enclosed Annexure XXXVIII
- Inspection certificates of BIU and test reports:
Enclosed Annexure XXXIX

6.2.3 Others

- Certificate from Shed that Kavach test yard is installed and routine maintenance of Kavach will be undertaken and use of only approved components in Kavach related loco/EMU/trainset interface:
Enclosed Annexure XL
- Jointly Signed pre commissioning check list:
Enclosed Annexure XXXII
- Approved fitment diagrams:
Enclosed Annexure XLII
- Approved circuit diagrams by the shed:
Enclosed Annexure XLIII
- OEM Installation certificate:
Enclosed Annexure XXXIII

6.2.4 KMS Registration details

Enclosed Annexure XLV

6.2.5 Training – Concerned Branch officer of shed shall certify that the staff are adequately trained for Kavach Operations and Maintenance.

Enclosed Annexure XLVI

6.2.6 Management of spares -

Enclosed Annexure XXXVII B

6.2.7 Maintenance staff nomination and posting.-

Enclosed Annexure XXXVII C

6.3 Towers

6.3.1 Tower location approval plan Tower location approved plan -

Enclosed Annexure XV

6.3.2 Soil Investigation Reports & recommendation for type of foundation:

Enclosed Annexure XIV

6.3.3 Details of the Approved tower foundations and structural drawings :

Enclosed Annexure XIII

6.3.4 Approved Quality Assurance Plan (QAP) for tower fabrication for 3rd party Inspection :

Enclosed Annexure L

6.3.5 3rd party Inspection Certificate for Tower fabrication (as applicable) :

Enclosed Annexure LI

6.3.6 Test reports for Cement, Sand, Aggregate M20, Steel and Cube test pertaining to tower foundations :

Enclosed Annexure LII

6.3.7 Stage-wise field inspection report for Tower foundation :

Enclosed Annexure LIII

6.3.8 Tower Verticality Test Report :

Enclosed Annexure LIV

6.3.9 Approved Cable route plan from Kavach to Tower location:

Enclosed Annexure LVII

6.3.10 Approved Cable Connectivity Diagram for Kavach to Tower :

Enclosed Annexure LVI

6.3.11 Approved cable route plan for Kavach Tower :

Enclosed Annexure LVII

6.3.12 Typical Earthing Plan for Kavach Tower:

Enclosed Annexure LVIII

6.3.13 Approved Tower fencing diagram:

Enclosed Annexure LIX

**6.3.14 Meggaring and OTDR details of cables from Kavach to Tower
[Clause 6.3.1 to 6.3.8 are applicable for new towers only] :**

Enclosed Annexure LX

6.4 Manuals

6.4.1 Stationary Kavach

- Maintenance check list:
Enclosed Annexure LXII
- Signed Pre-commissioning check list:
Enclosed Annexure XXXII
- Configuration manuals:
Enclosed User Manual for configuration tools for Kavach v2.0 : Enclosed Annexure LXVIII

6.4.2 Onboard Kavach

- Maintenance check list:
Enclosed Annexure LXVI
- Signed Pre-commissioning check list:
Enclosed Annexure LXVII
- Configuration manuals:
Enclosed Annexure LXVIII

6.4.3 TSRMS –

Not Applicable

- Maintenance check list:
- Signed Pre-commissioning check list:
- Configuration manuals:

7 ISA Report

7.1 About ISA

Name of ISA: ITALCERTIFER Enclosed Annexure LXIX

7.2 OEM's organization structure

(Enclosed Annexure LXX)

7.3 Fulfillment of Safety Related Application conditions mentioned in the generic safety case

HBL's document # 5 51 53 0010 for Fulfillment of SRACs as enclosed Annexure LXXI

7.4 Specific Application Safety Case (SASC) requirements of KAVACH deployed in section

HBL's document # 5 50 53 0006 for SASC is enclosed as Annexure LXXII.

7.5 Hazard log and mitigation measures

HBL's document # 5 50 53 0003 for Hazard log and mitigation measures as enclosed Annexure LXXIII

7.6 ISA Field audit report

ITCF Test witnessing report # ITCF-C25020-DT-CCS-07-I-00-00-TE-P001 dated 29th Apr 2025 is enclosed as Annexure LXXIV

7.7 Final ISA assessment Report

ITCF's ISA report # ITCF-C25020-DT-CCS-09-I-00-00-TE-P001 dated 02nd July 2025 is enclosed as Annexure LXXV

8 Performance Report

8.1 Daily Incidences summary for 30 days

(Enclosed Annexure LXXVI)

8.2 Operational Availability (minimum 98.5%)

(Enclosed Annexure LXXVI)

NOTE: This report is system generated and doesn't need any physical signature